

1. Dataset Overview

- **Total records:** 1,297,015
- **Columns:** 35
- **Sites:** 106

dataset	rows	sites	sampling	first_ts	last_ts
goes_pvdaq	88806	8	15 minutes	2019-01-01 00:00:00+00:00	2019-09-30 23:45:00+00:00
uk_pv	1208209	98	30 minutes	2019-01-01 08:00:00+00:00	2020-12-31 16:00:00+00:00

1.1 Organization

Two source datasets (uk_pv satellite HRV crops, goes_pvdaq GOES-16 crops) merged into a single table `dataset_all.parquet` — one row per (`site_id`, `timestamp_utc`). Each row pairs a PV power reading and its weather/solar covariates with a co-registered image frame; pixels live in `images_all.h5`, referenced by the integer `image*_index` pointers. `norm_power` ($\text{power} \div \text{installed capacity}$) is the model target; splits are cross-plant on `site_id`.

1.2 Columns

Column	Meaning	Role
<code>dataset</code>	Source dataset: <code>uk_pv</code> or <code>goes_pvdaq</code>	id — cross-region split key
<code>site_id</code>	PV plant identifier	id — cross-plant split key
<code>timestamp_utc</code>	UTC sample time	index — windowing
<code>power_w</code>	Measured PV power output (W)	raw target; not fed directly
<code>installed_power_w</code>	Nameplate capacity (W)	normalizer
<code>norm_power</code>	$\text{power_w} \div \text{installed_power_w}$	target
<code>temperature_2m</code>	2 m air temperature (°C)	covariate — observed
<code>shortwave_radiation</code>	Surface downwelling shortwave (W/m ²)	covariate — observed
<code>direct_radiation</code>	Direct-beam component (W/m ²)	covariate — observed
<code>diffuse_radiation</code>	Diffuse component (W/m ²)	covariate — observed
<code>direct_normal_irradiance</code>	DNI (W/m ²)	covariate — observed
<code>cloudcover</code>	Total cloud cover (%)	covariate — observed
<code>windspeed_10m</code>	10 m wind speed (m/s)	covariate — observed
<code>precipitation</code>	Precipitation (mm)	covariate — observed
<code>solar_zenith</code>	Solar zenith angle (deg)	covariate — deterministic
<code>solar_azimuth</code>	Solar azimuth angle (deg)	covariate — deterministic
<code>clearsky_ghi</code>	Clear-sky GHI model (W/m ²)	covariate — deterministic
<code>doy_sin, doy_cos</code>	Cyclic day-of-year encoding	covariate — deterministic
<code>solar_time</code>	Local apparent solar time (hours)	covariate — deterministic
<code>image_h5_index</code>	Row index into <code>images_all.h5</code>	vision input pointer
<code>latitude, longitude</code>	Site coordinates	metadata — unused (can be used for graph building)
<code>station_id</code>	Weather station tag	metadata — unused
<code>camera_id</code>	Imagery channel (HRV/GOES)	metadata — unused
<code>outage_flag</code>	Flag: outage / zero-output anomaly	curation only, unused
<code>night_clamped</code>	Flag: nighttime power clamped to 0	curation only, unused

1.3 Construction & sources

The dataset is assembled fusing four public sources into one standardized (`site_id`, `timestamp_utc`) table plus co-registered image frames:

- **PV power — UK** ([openclimatefix/uk_pv](#), HF): 30-minutely generation (2019–2020) and per-site metadata (rounded lat/lon, kWp capacity). `generation_Wh` over 30 min is converted to watts ($\times 2$); short gaps (≤ 3 steps) are linearly interpolated, unfixable ones dropped.
- **UK satellite imagery — EUMETSAT SEVIRI RSS HRV** from the Open Climate Fix public GCS bucket (`gs://public-datasets-eumetsat-solar-forecasting/satellite/EUMETSAT/SEVIRI_RSS/v4/<year>_hrv.zarr`), ~ 1 km/px. Frames are reprojected (pyproj) to each site and cropped to **128×128 px** (~ 128 km) for cloud-advection context.
- **PV power + imagery — US** (`goes_pvdaq`): NREL **PVDAQ** systems with site metadata (lat/lon, nameplate power) from the [OEDI data lake](#), paired with **GOES-16** satellite crops (10 sites).
- **Weather covariates — Open-Meteo Archive API**: 8 hourly variables (2 m temperature, shortwave/direct/diffuse radiation, DNI, cloud cover, 10 m wind speed, precipitation), fetched per site and joined to each row by nearest-timestamp (`merge_asof`).

Operations. Per frame: NaN $\rightarrow$ 0, clip to [0,1], keep only non-empty crops; images are written as per-site PNGs / a single uint8 HDF5 (millions of small files are impractical). Numeric rows from both sources are concatenated, enriched with `installed_power_w` + coordinates, then joined with the weather track and saved.

1.4 Other public dataset — ClimateHackAI 2023

The **ClimateHackAI 2023** competition dataset ([DOXA · HF](#), ~ 600 GB) is a multimodal PV-nowcasting corpus over **Great Britain**. Per site it provides **PV generation** as a fraction of installed capacity at **5-min** resolution, with metadata (latitude, longitude, orientation, tilt, installed capacity); two **EUMETSAT** satellite streams at 5-min cadence — **HRV** [12,128,128] and **non-HRV** 11-channel [12,128,128,11] crops centred on each site; **DWD ICON-EU** numerical weather forecasts (38 variables, T–1h...T+4h, 128×128 grids); and **ECMWF CAMS** air-quality forecasts (13 variables $\times$ 8 altitudes). The competition task: from **1 h of history** (PV + satellite) plus the hourly weather/air-quality forecast steps, predict the **next 4 h** of PV output — 48 values at 5-min intervals — scored by mean absolute error (target < 0.15).

It is structured differently from the dataset of record: built for **nowcasting**, it is organised as short fixed windows (1 h in $\rightarrow$ 4 h out) at 5-min resolution with per-window satellite/NWP/air-quality tensors, rather than a single long tall table of (`site`, `timestamp`) rows spanning history and horizon. That short-window, high-cadence, forecast-tensor layout targets minutes-to-hours nowcasting and would need re-windowing and re-sampling before it could serve the longer-history, cross-plant protocol used here.

Note: this is a **competition dataset with no accompanying published paper**; it is documented only through the competition overview and Hugging Face repository, so there is no peer-reviewed reference to cite for its construction or provenance.

1.5 Other public dataset — MMSP (FusionSF)

The **MMSP** (Multi-Modal Solar Power) dataset is released with **FusionSF** ([arXiv:2402.05823 · code · data](#)), a multimodal PV corpus over **88 geographically dispersed solar plants across a Chinese province** ($\sim 157,100$ km²), spanning **Jan 2021 – Jun 2022** at **hourly** resolution (downsampled from 10-min). Each record fuses three modalities: **historical PV power**; **Himawari-8/9 satellite imagery** cropped to **64×64 px** (from 640×640) — 1 channel in the small variant **MMSP(S)** (10 plants), 4 visible/near-infrared channels in the full **MMSP(L)** (88 plants); and **ECMWF NWP** (17 weather features: radiation, temperature, pressure, cloud cover, wind). The task is **day-ahead** forecasting: 24 h of history $\rightarrow$ next 24 h. (The paper reports the method deployed on 300+ plants / >15 GW in production; MMSP is the released research subset.)

Like ClimateHackAI, it differs structurally from the dataset of record — hourly cadence, fixed day-ahead 24-in $\rightarrow$ 24-out windows over China with Himawari imagery and per-window NWP tensors — so it would need re-windowing/re-sampling before serving the longer-history, cross-plant protocol used here.

2. Data Splits

Cross-plant, seeded per-dataset partition over the 106 sites. Counts combine both datasets.

	Plants	Rows
Train	75	918 158
Validation	16	193 553
Test	15	185 304

Per-dataset plants — uk_pv: 69/15/14 (train/val/test); goes_pvdaq: 6/1/1.

3. Configurations

Dimension	Specs
History	14 days (steps: 672 uk and 1344 pvdaq)
Horizon	6 hours (steps: 12 uk and 24 pvdaq)
Frames	8 frames @ 60-min cadence (last 8 h ending at forecast time)

Models should be tested also at intra-hour frame cadence. On uk_pv 30 min and on goes_pvdaq 15 min can be used (longer training).

2. Leaderboard

Rank is the global position by skill score (SS $\uparrow$) across all models; each subtable is sorted best-first. Metrics: SS $\uparrow$, R^2 $\uparrow$, CRPS $\downarrow$. Reference floor is **smart_persistence** (SS $\equiv$ 0).

Metric intuition. All point metrics run on the capacity-normalized target (**norm_power** $\in$ [0,1]) over daylight, valid steps, so values read as fraction-of-nameplate error and compare across plants.

- **R^2 $\uparrow$** — squared Pearson correlation between predicted and true power, $r^2 = \text{corr}(\text{true}, \text{pred})^2$, pooled across test plants on daylight, valid steps. Measures how much of the *variance* the forecast tracks (shape/timing), independent of bias or scale. 1 = perfectly correlated, 0 = no linear relation.
- **SS (skill score) $\uparrow$** — how much a model beats the **smart_persistence** baseline on root-mean-square error: 0 = no better than baseline, 1 = perfect, negative = worse.
- **CRPS $\downarrow$** — scores the *whole predictive distribution* (via quantile pinball), not just the mean; rewards forecasts both accurate *and* honestly calibrated about uncertainty (probabilistic models only).

Cov flag. Each table carries a **Cov** column: $\checkmark$ = the model ingests the weather/solar covariate track

2.1 Reference / statistical

Rank	Model	SS $\uparrow$	R^2 $\uparrow$	CRPS $\downarrow$	Cov	Description
23	climatology_ hourly	0.234	0.314	—	$\times$	Per-hour-of-day mean power from the training set; a fixed seasonal climatology.
28	seasonal_naive	0.107	0.276	—	$\times$	Copies power from the same clock time on the previous day.
30	persistence	0.014	0.171	—	$\times$	Carries the last observed power flat across the horizon.
—	smart_ persistence	0.000	0.210	—	$\times$	Persistence rescaled by the clear-sky ratio — the sun-adjusted reference floor

2.2 Classical ML

Rank	Model	SS $\uparrow$	R ² $\uparrow$	CRPS $\downarrow$	Cov	Description
11	LightGBM	0.384	0.555	0.0768	✓	Gradient-boosted decision trees over lagged power + covariate features.

2.3 Tabular foundation models

Rank	Model	SS $\uparrow$	R ² $\uparrow$	CRPS $\downarrow$	Cov	Paper
17	TabPFN 3	0.339	0.499	0.0815	✓	arXiv:2605.13986
—	TabFM 1.0	TBD	—	TBD	✓	Google Research

2.4 Supervised DL

Rank	Model	SS $\uparrow$	R ² $\uparrow$	CRPS $\downarrow$	Cov	Description
1	iTransformer	0.552	0.765	—	✓	arXiv:2310.06625
7	PatchTST	0.458	0.651	—	✓	Splits each series into patches for a channel-independent transformer.
9	TFT	0.423	0.626	0.0689	✓	Temporal Fusion Transformer; gated attention with quantile outputs.
10	MLP	0.413	0.600	—	✓	Plain multilayer perceptron over the flattened history window.
20	DLinear	0.325	0.462	—	×	Single linear layer on trend/seasonal decomposition — target-only simplicity check.

2.5 Zero-shot TSFM

Rank	Model	SS $\uparrow$	R ² $\uparrow$	CRPS $\downarrow$	Cov	Paper
21	tirex_zs	0.287	0.440	0.0892	×	arXiv:2505.23719
22	timesfm_zs	0.271	0.429	0.0923	×	TimesFM 2.5
26	chronos2_zs	0.187	0.277	0.1072	✓	arXiv:2510.15821
×	ttn_zs	−0.081	0.155	—	✓	TTM-R2

2.6 Fine-tuned TSFM

Rank	Model	SS $\uparrow$	R ² $\uparrow$	CRPS $\downarrow$	Cov	Paper
3	chronos2_oracle_ft	0.504	0.706	0.0630	✓	arXiv:2510.15821
6	chronos2_oracle	0.474	0.679	0.0635	✓	arXiv:2510.15821
13	ttn_ft	0.364	0.520	—	✓	TTM-R2
19	chronos2_ft	0.331	0.475	0.0855	✓	arXiv:2510.15821

chronos2_* vs chronos2_oracle_* — same Chronos-2 model; the only difference is *which covariates are exposed over the forecast horizon*. The plain variants (**chronos2_zs**, **chronos2_ft**) uses only deterministic future covariates (solar geometry + calendar) that are known ahead. The **oracle** variants feed the model the **future observed weather** (temperature, cloud cover, irradiance, ...) over the horizon.

2.7 Retrieval / frozen-FM adaptation

Rank	Model	SS $\uparrow$	R ² $\uparrow$	CRPS $\downarrow$	Cov	Regime	Paper
4	ts_rag	0.478	0.331	—	×	Zero-shot (frozen FM + retrieval)	arXiv:2503.07649
5	cross_rag	0.477	0.375	—	×	Zero-shot (frozen FM + retrieval)	arXiv:2603.14709
12	cora	0.374	0.550	0.0816	✓	Fine-tuned (adapter, frozen FM)	arXiv:2510.12681

2.8 Endogenous multimodal (second modality synthesized from the numeric track — no external sensor)

Pseudo-images rendered from the series (time_vlm, visions_pp) or weather-text templated from the covariates (aurora). No satellite/sky frames consumed.

Rank	Model	SS $\uparrow$	R ² $\uparrow$	CRPS $\downarrow$	Cov	Regime	Paper
2	time_vlm	0.540	0.487	—	×	Fine-tuned (frozen VLM backbones)	arXiv:2502.04395
24	aurora	0.232	0.440	—	✓	Zero-shot	arXiv:2509.22295
29	visions_pp	0.017	0.397	—	×	Zero-shot (frozen image MAE)	arXiv:2508.04379

2.9 Exogenous multimodal (external satellite imagery)

Rank	Model	SS $\uparrow$	R ² $\uparrow$	CRPS $\downarrow$	Cov	Regime	Paper
8	solar_vlm	0.443	0.660	—	✓	Fine-tuned (pretrained vision)	arXiv:2604.04145
14	crossvivit	0.349	0.565	—	✓	From scratch	arXiv:2306.01112
16	mmtsfm__ grassmann__ interleaved	0.3429	—	0.0805	✓	Fine-tuned (frozen FM + vision)	ours (this work)
25	sunset	0.216	0.359	—	×	From scratch	github: YuchiSun/SUNSET
27	unicast	0.121	0.371	—	×	Fine-tuned (soft-prompt, frozen FM)	arXiv:2508.11954
—	FusionSF	TBD	—	TBD	✓	TBD	arXiv:2402.05823
—	PIPE	TBD	—	TBD	✓	TBD	arXiv:2506.14786
—	MATE	TBD	—	TBD	✓	TBD	OpenReview

4. Conclusions

Drawn from the cross-plant leaderboard. All numbers are zero-shot on **disjoint test plants** — no test-plant data in training.

1. A well-tuned supervised transformer is the model to beat. iTransformer tops every metric (SS 0.552, R^2 0.765), ahead of every foundation model, retrieval scheme, and multimodal system. On this cross-plant PV task, architecture + in-domain supervised training still outperforms large pretrained backbones.

2. Zero-shot time-series foundation models underdeliver on PV. The zero-shot TSFMs (chronos2_zs SS 0.187, timesfm_zs 0.271, tirex_zs 0.287, ttm_zs **-0.081** — worse than the naive floor) trail even classical baselines. PV’s sharp, weather-driven dynamics are out-of-distribution for generic TS pretraining; the models need adaptation to be useful.

3. Adaptation closes the gap, and retrieval is the best frozen-backbone lever. Every adaptation step helps: Chronos-2 jumps zero-shot $\rightarrow$ fine-tuned, and TTM flips from below-baseline to SS 0.364 once fine-tuned. Notably, **retrieval over a frozen backbone** (ts_rag SS 0.478, cross_rag 0.477) rivals fine-tuning **without updating any weights**, making it the most effective — and cheapest — frozen-FM strategy tested.

4. Multimodality has not yet paid off The one multimodal model at the top (time_vlm, SS 0.540, rank 2) uses *endogenous pseudo-images synthesized from the numeric series* and no covariates. Every model that ingests genuine **exogenous satellite/sky imagery** ranks mid-pack or worse.

7. R^2 and SS disagree for biased forecasters — read them together. Several models track the day’s *shape* well yet sit off the diagonal: e.g. visions_pp has R^2 0.397 but SS 0.017, and solar_vlm / sunset / crossvivit show systematic under-prediction. High variance-tracking (R^2) with low skill (SS) flags a **bias/scale** problem, not a timing problem — fixable by calibration rather than a better temporal model.